

RAG Based questions

Q1. What is the importance of chunking in RAG? What are the consequences of chunks that are too large vs. too small?

Answer

Importance of Chunking

Chunking is the process of splitting large documents into smaller segments before generating embeddings and storing them for retrieval. It improves retrieval accuracy because the system can find specific relevant information instead of retrieving entire documents.

Benefits include:

- Improves retrieval precision
- Helps embeddings capture clear semantic meaning
- Reduces irrelevant text sent to the LLM
- Helps manage LLM context window limits

Chunks that are Too Large

If chunks are too large:

- Embeddings may represent multiple topics
- Retrieval may include irrelevant information
- LLM context window may be wasted

Example: A chunk containing installation, troubleshooting, and performance tuning together may reduce retrieval accuracy.

Chunks that are Too Small

If chunks are too small:

- Important context may be lost
- Retrieved text may be incomplete
- More chunks may need to be retrieved

Example: "The server failed during deployment" lacks enough context to explain the issue.

Recommended Chunk Size

Typical configuration:

- 200–500 tokens per chunk
- 20–50 token overlap

Q2. Compare keyword-based retrieval vs semantic retrieval in a RAG system. When would you choose one over the other?

Answer

Keyword-Based Retrieval

Keyword retrieval finds documents containing exact query terms.

Tools used:

- BM25
- Elasticsearch
- Lucene

Advantages:

- Fast retrieval
- Works well with IDs, codes, and specific terms

Example:

Query: “Error code ERR5042”

Semantic Retrieval

Semantic retrieval searches based on meaning using embeddings.

Tools used:

- Vector databases like FAISS, Pinecone, Milvus

Advantages:

- Understands natural language
- Handles synonyms and paraphrasing

Example:

Query: “How can I improve database performance?”

When to Use Each

Keyword retrieval is useful for:

- product IDs
- error codes
- exact terms

Semantic retrieval is useful for:

- natural language queries
- conceptual questions

Best practice: use hybrid retrieval combining both methods.

Q3. What are the pros and cons of using re-rankers in RAG? How do they affect latency vs accuracy?

Answer

Re-rankers reorder retrieved documents based on their relevance to the query.

Advantages

- Improves retrieval accuracy
- Places the most relevant documents at the top
- Reduces irrelevant results

Disadvantages

- Increases system latency
- Requires additional computation

Latency vs Accuracy

Typical pipeline:

Retrieve Top 50 Documents

↓

Re-rank Documents

↓

Select Top 5

↓

Send to LLM

Re-rankers improve accuracy, but they increase response time.

Q4. How would you scale embeddings in a RAG retrieval system to handle billions of documents?

Answer

Large RAG systems require scalable vector search techniques.

Key Methods

Approximate Nearest Neighbor (ANN)

Algorithms such as:

- HNSW
- IVF
- ScaNN

These allow fast similarity search in very large datasets.

Vector Database Sharding

Indexes are distributed across multiple servers.

Example:

Server 1 → Documents 1–100M

Server 2 → Documents 100–200M

Hierarchical Retrieval

Two-step retrieval:

1. Retrieve document clusters
2. Search within clusters

This reduces search time.

Metadata Filtering

Filtering by attributes like:

- document type
- department
- timestamp

helps narrow the search space.

Q5. How do you handle documents that exceed the LLM context window in a RAG system?

Answer

Large documents often exceed the context window of language models. Several strategies are used to handle this.

Chunk-Based Retrieval

Split documents into chunks and retrieve only the most relevant ones.

Hierarchical Summarization

Create summaries in stages: Chunks → Section summaries → Final summary

Sliding Window Retrieval

Retrieve overlapping sections of long documents and select the most relevant parts.

Graph-Based Retrieval

Use knowledge graphs to retrieve only relevant entities and relationships instead of the entire document.

Benefits

- Efficient use of context window
- Faster responses
- Improved answer accuracy